LESSON 4-4
Adding and Subtracting Rational Expressions

Lesson Overview

FOCUS

Objective
Students will be able to:

✔ Understand that rational expressions form a system analogous to the system of rational numbers and use that understanding to add and subtract rational expressions.

Essential Understanding
The properties of operations used to add and subtract rational numbers can be applied to adding and subtracting rational expressions.

COHERENCE

Previously in this course, students:
- Added and subtracted rational numbers.
- Multiplied and divided rational expressions using methods similar to those used with rational numbers.

In this lesson, students:
- Use their understanding of operations with rational numbers to add and subtract rational expressions.

Increasing Coherence Examples 3–6 are not necessarily expected to be addressed on high stakes assessments.

Later in this topic, students will:
- Add, subtract, multiply, and divide rational expressions in the context of solving rational equations.

RIGOR

This lesson emphasizes a blend of *procedural skill and fluency* and *application.*
- Students identify the least common denominator of two rational expressions and use it to add and subtract the expressions.
- Students apply adding and subtracting rational expressions to solve problems involving rate and time.

Glossary is available at SavvasRealize.com.

Vocabulary Builder

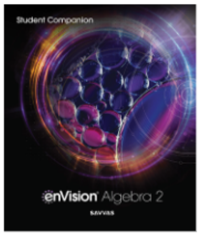
REVIEW VOCABULARY
English | *Spanish*
- **algebraic fraction** | *fracción algebraica*
- **numeric fraction** | *fracción numérica*

NEW VOCABULARY
- **compound fraction** | *fración compuesta*

VOCABULARY ACTIVITY
Review the term *fraction* with students and have them differentiate between *algebraic fractions* and *numeric fractions.* Next have students consider the word *compound,* and use the definition of that to infer the meaning of *compound fraction* as it applies to algebraic fractions. Have students use a Venn Diagram to sort the fractions shown below to identify which are numeric, which are algebraic, and which are compound.

$$\frac{4}{5} \qquad \frac{x+4}{5} \qquad \frac{\frac{3+n}{4-n}}{6-n} \qquad \frac{x^2}{(x+2)} \qquad \frac{\frac{1}{2}}{6} \qquad \frac{\frac{(x+2)^2}{(x-4)}}{\frac{(x+6)}{(x-3)}} \qquad \frac{3}{4+5}$$

Student Companion
Students can do their work for the lesson in their *Student Companion* or on Savvas Realize.



Mathematics Overview
Students use their knowledge of adding and subtracting rational numbers to add and subtract rational expressions They first rewrite each expression in terms of a common denominator, then add or subtract the numerators.

Applying Math Practices
Use Appropriate Tools Strategically
Students use mathematical models, such as tables, to visualize and analyze information presented in a problem about finding rate.

Look for and Make Use of Structure
Students see algebraic fractions as being composed of parts that are similar to rational numbers and recognize that this allows them to add and subtract rational expressions in the same way as rational numbers.

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Activity

CRITIQUE & EXPLAIN

INSTRUCTIONAL FOCUS In preparation for adding and subtracting rational expressions, students use their knowledge of common denominators to find a method for adding fractions with unlike denominators.

STUDENT COMPANION Students can complete the *Critique & Explain* activity in their *Student Companion.*

Before WHOLE CLASS

Implement Tasks that Promote Reasoning and Problem Solving ETP

Q: What do you notice about the expression in the homework exercise? [Three fractions are added. The numerator of each fraction is 1. Each fraction has a different denominator.]

During SMALL GROUP

Support Productive Struggle in Learning Mathematics ETP

Q: Compare common multiples and the least common multiple. How are they similar? How are they different?
[The least common multiple is a common multiple. The common multiples of a group of numbers are multiples of each number in the group. A group of numbers has many common multiples, but only one least common multiple.]

For Early Finishers

Q: Add $\frac{1}{8}$ to the expression in the homework exercise. What are the least common denominator and the sum of the new expression? [72; $\frac{77}{72}$]

After WHOLE CLASS

Facilitate Meaningful Mathematical Discourse ETP

Q: Why would you want to use the least common multiple when writing fractions with a common denominator instead of any other common multiple?
[The least common multiple gives you smaller numbers to compute.]

Q: How can there be more than one correct answer to this problem?
[You can write the answer with different equivalent fractions. All of these answers to the homework exercise are equivalent: $\frac{17}{18}$, $\frac{34}{36}$, $\frac{51}{54}$, and $\frac{68}{72}$.]

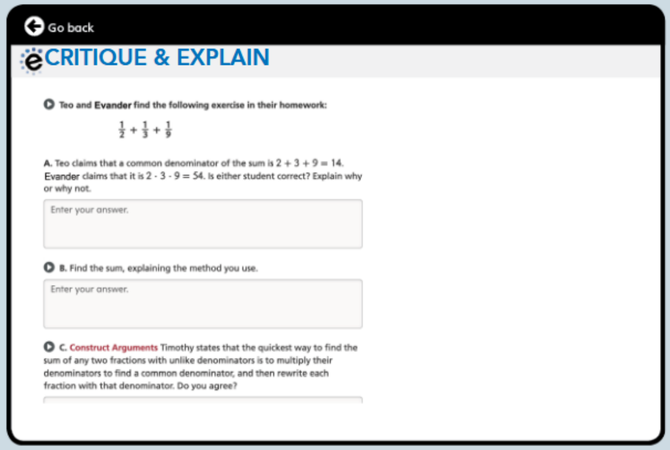
Use with CRITIQUE & EXPLAIN

HABITS OF MIND

Look for Relationships For two fractions with denominators 10 and x, when is $10x$ the least common multiple? When is $10x$ NOT the least common multiple?

[When x and 10 have no common factors, $10x$ is the least common multiple; when x and 10 have common factors, $10x$ is not the least common multiple.]

Model & Discuss is available at SavvasRealize.com.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CRITIQUE & EXPLAIN

Teo and Evander find the following exercise in their homework:

$\frac{1}{2}+\frac{1}{3}+\frac{1}{9}$

A. Teo claims that a common denominator of the sum is $2+3+9=14$. Evander claims that it is $2 \cdot 3 \cdot 9 = 54$. Is either student correct? Explain why or why not.

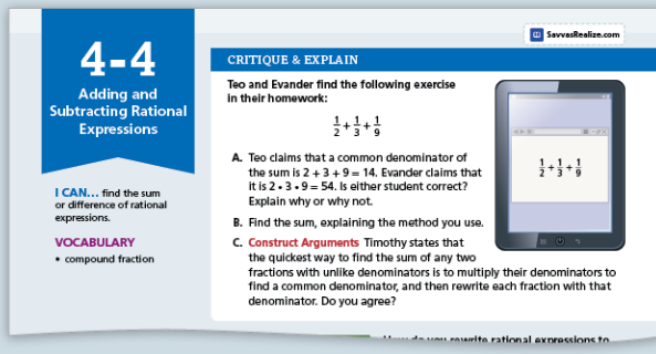
Enter your answer.

B. Find the sum, explaining the method you use.

Enter your answer.

C. Construct Arguments Timothy states that the quickest way to find the sum of any two fractions with unlike denominators is to multiply their denominators to find a common denominator, and then rewrite each fraction with that denominator. Do you agree?

STUDENT EDITION


4-4
Adding and Subtracting Rational Expressions

I CAN… find the sum or difference of rational expressions.

VOCABULARY
• compound fraction

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CRITIQUE & EXPLAIN

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SAMPLE STUDENT WORK

A. Only Evander is correct. A common denominator will have each other denominator as a factor.

B. $\frac{17}{18}$; converted each fraction to a fraction with denominator 18.

C. No; It may be faster to use the least common denominator rather than multiplying the denominators together.

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Activity

INTRODUCE THE ESSENTIAL QUESTION

Establish Mathematics Goals to Focus Learning ETP

Introduce students to adding and subtracting rational expressions. Explain that the rules for adding and subtracting rational expressions are similar to the rules for adding and subtracting fractions.

EXAMPLE 1 Add Rational Expressions With Like Denominators

Pose Purposeful Questions ETP

Q: What do you notice about the denominators in Part A?
[The denominators of the rational expressions are the same.]

Q: What do you notice about the denominators in Part B?
[Even though the denominators are written differently, they are equivalent: $x^2 + 3x = x(x + 3)$.]

Try It! Answers

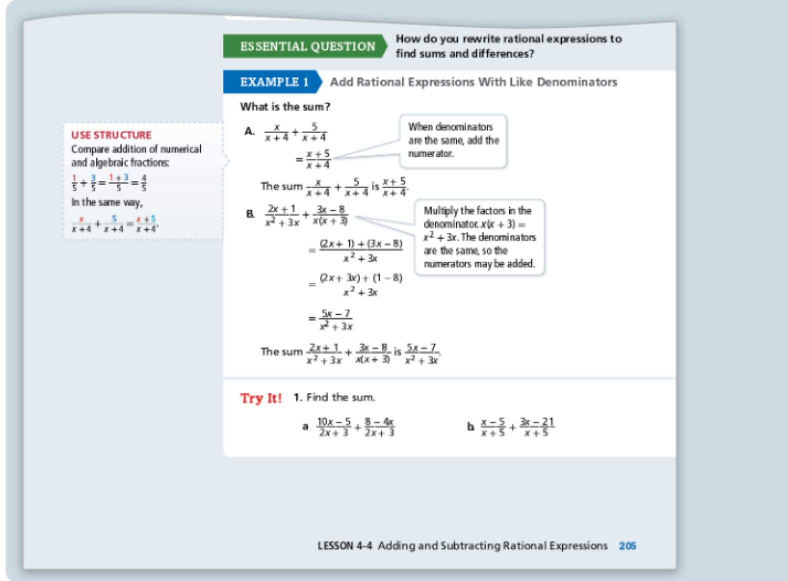
1. a. $\dfrac{6x+3}{2x+3}$ or $\dfrac{3(2x+1)}{2x+3}$

   b. $\dfrac{4x-26}{x+5}$ or $\dfrac{2(2x-13)}{x+5}$

Elicit and Use Evidence of Student Thinking ETP

Q: Can you add the numerators of these rational expressions?
[Yes; the denominators of the rational expressions are the same, so you can add the numerators.]

Examples are available at SavvasRealize.com.



ESSENTIAL QUESTION How do you rewrite rational expressions to find sums and differences?

EXAMPLE 1 Add Rational Expressions With Like Denominators

What is the sum?

A. $\dfrac{x}{x+4} + \dfrac{5}{x+4}$

$= \dfrac{x+5}{x+4}$

When denominators are the same, add the numerator.

The sum $\dfrac{x}{x+4} + \dfrac{5}{x+4}$ is $\dfrac{x+5}{x+4}$.

B. $\dfrac{2x+1}{x^2+3x} + \dfrac{3x-8}{x(x+3)}$

$= \dfrac{(2x+1)+(3x-8)}{x^2+3x}$

$= \dfrac{(2x+3x)+(1-8)}{x^2+3x}$

$= \dfrac{5x-7}{x^2+3x}$

Multiply the factors in the denominator. $x(x+3) = x^2 + 3x$. The denominators are the same, so the numerators may be added.

The sum $\dfrac{2x+1}{x^2+3x} + \dfrac{3x-8}{x(x+3)}$ is $\dfrac{5x-7}{x^2+3x}$.

USE STRUCTURE
Compare addition of numerical and algebraic fractions:
$\frac{1}{5} + \frac{3}{5} = \frac{1+3}{5} = \frac{4}{5}$
In the same way,
$\frac{x}{x+4} + \frac{5}{x+4} = \frac{x+5}{x+4}$.

Try It! 1. Find the sum.

a. $\dfrac{10x-5}{2x+3} + \dfrac{8-4x}{2x+3}$

b. $\dfrac{x-5}{x+5} + \dfrac{3x-21}{x+5}$

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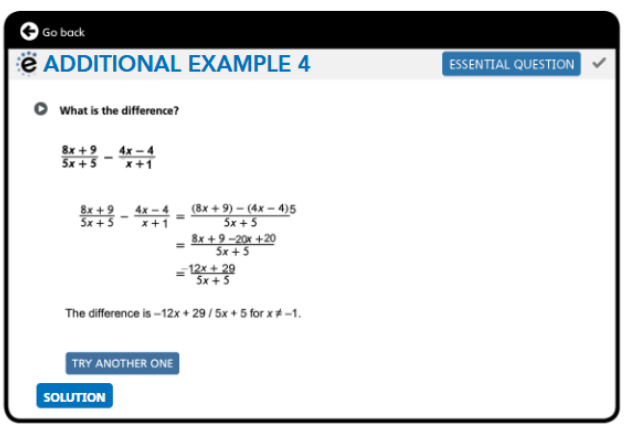
HABITS OF MIND

Use with EXAMPLE 1

Make Sense and Persevere Explain why it does not make sense to add the denominators when adding rational numbers. Use numerical fractions to support your thinking.

[When adding two fractions such as $\frac{1}{5} + \frac{2}{5}$, you add their numerators and keep the common denominator to get a sum of $\frac{3}{5}$. You use the same process to add rational numbers.]

Additional Examples



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ADDITIONAL EXAMPLE 4 ESSENTIAL QUESTION

What is the difference?

$\dfrac{8x+9}{5x+5} - \dfrac{4x-4}{x+1}$

$\dfrac{8x+9}{5x+5} - \dfrac{4x-4}{x+1} = \dfrac{(8x+9)-(4x-4)5}{5x+5}$

$= \dfrac{8x+9-20x+20}{5x+5}$

$= \dfrac{-12x+29}{5x+5}$

The difference is $-12x + 29 / 5x + 5$ for $x \neq -1$.

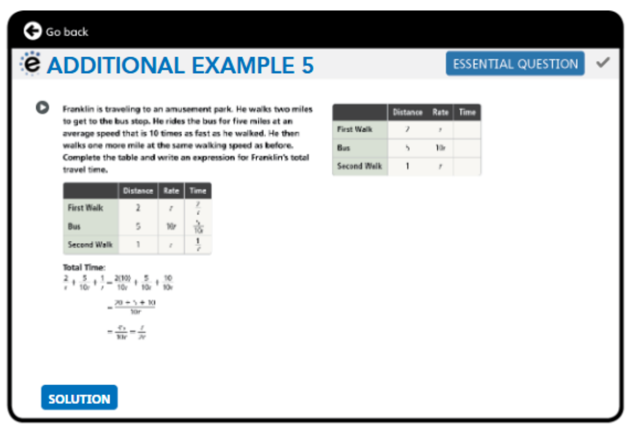
TRY ANOTHER ONE

SOLUTION

Example 4 Help students understand adding and subtracting rational expressions with unlike denominators.

Q: What values are not in the domain when adding or subtracting rational expressions?
[Any value that makes any denominator equal to 0 is not in the domain.]

Additional Examples are available at SavvasRealize.com.



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ADDITIONAL EXAMPLE 5 ESSENTIAL QUESTION

Franklin is traveling to an amusement park. He walks two miles to get to the bus stop. He rides the bus for five miles at an average speed that is 10 times as fast as he walked. He then walks one more mile at the same walking speed as before. Complete the table and write an expression for Franklin's total travel time.

	Distance	Rate	Time
First Walk	2	r	
Bus	5	$10r$	
Second Walk	1	r	

	Distance	Rate	Time
First Walk	2	r	$\frac{2}{r}$
Bus	5	$10r$	$\frac{5}{10r}$
Second Walk	1	r	$\frac{1}{r}$

Total Time:

$\frac{2}{r} + \frac{5}{10r} + \frac{1}{r} = \frac{2(10)}{10r} + \frac{5}{10r} + \frac{10}{10r}$

$= \frac{20+5+10}{10r}$

SOLUTION

Example 5 Help students transition to solving a rate problem for a three-part trip.

Q: Can you determine the time of the trip with the information given? Explain. [No; you need to know the speed that Franklin walked.]

EXAMPLE 2 Identify the Least Common Multiple of Polynomials

Build Procedural Fluency from Conceptual Understanding ETP

Q: How can you find the LCM of two rational expressions if the polynomials cannot be factored?
[You multiply the two polynomials.]

Try It! Answers

2. a. $(x + 3)^3 (x - 7)$ b. $30(x + y)(x + 2y)(x - y)^2$

Elicit and Use Evidence of Student Thinking ETP

Q: How could the LCM of two polynomials be one of the polynomials?
[One of the polynomials is a factor of the other.]

EXAMPLE 3 Add Rational Expressions With Unlike Denominators

Pose Purposeful Questions ETP

Q: Why would you want to find the sum using LCM instead of any other common multiple?
[Using the LCM will reduce the amount of simplifying you need to do after adding.]

Q: How is thinking about adding fractions with unlike denominators helpful when adding rational expressions with unlike denominators?
[The process that you use to add fractions with unlike denominators is similar to the process for adding rational expressions with unlike denominators. Thinking about fractions may help simplify the process.]

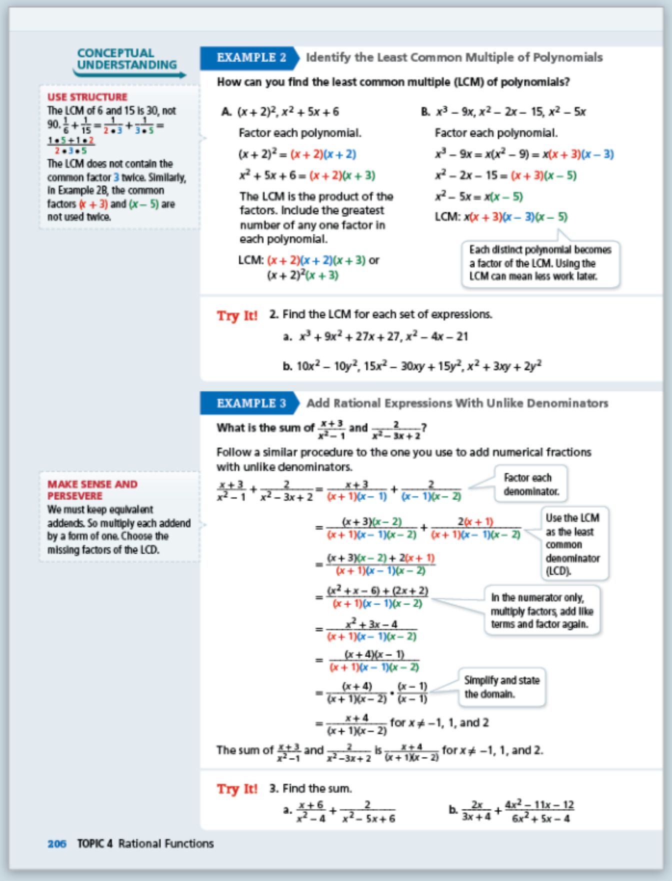
Try It! Answers

3. a. $\dfrac{x + 7}{(x + 2)(x - 3)}$ for $x \neq -2$, 2, and 3.
b. $\dfrac{8x^2 - 13x - 12}{(2x - 1)(3x + 4)}$ for $x \neq -\frac{4}{3}$ and $\frac{1}{2}$.

Elicit and Use Evidence of Student Thinking ETP

Q: Why should you try to factor the expression after adding the numerators?
[You factor the expression to see if there are any common factors you can divide out of the numerator and denominator.]

Examples are available at SavvasRealize.com.



CONCEPTUAL UNDERSTANDING

EXAMPLE 2 Identify the Least Common Multiple of Polynomials

How can you find the least common multiple (LCM) of polynomials?

USE STRUCTURE
The LCM of 6 and 15 is 30, not 90. $\frac{1}{6} + \frac{1}{15} = \frac{1}{2 \bullet 3} + \frac{1}{3 \bullet 5} = \frac{1 \bullet 5 + 1 \bullet 2}{2 \bullet 3 \bullet 5}$
The LCM does not contain the common factor 3 twice. Similarly, in Example 2B, the common factors $(x + 3)$ and $(x - 5)$ are not used twice.

A. $(x + 2)^2, x^2 + 5x + 6$
Factor each polynomial.
$(x + 2)^2 = (x + 2)(x + 2)$
$x^2 + 5x + 6 = (x + 2)(x + 3)$
The LCM is the product of the factors. Include the greatest number of any one factor in each polynomial.
LCM: $(x + 2)(x + 2)(x + 3)$ or $(x + 2)^2(x + 3)$

B. $x^3 - 9x, x^2 - 2x - 15, x^2 - 5x$
Factor each polynomial.
$x^3 - 9x = x(x^2 - 9) = x(x + 3)(x - 3)$
$x^2 - 2x - 15 = (x + 3)(x - 5)$
$x^2 - 5x = x(x - 5)$
LCM: $x(x + 3)(x - 3)(x - 5)$

Each distinct polynomial becomes a factor of the LCM. Using the LCM can mean less work later.

Try It! 2. Find the LCM for each set of expressions.
a. $x^3 + 9x^2 + 27x + 27, x^2 - 4x - 21$
b. $10x^2 - 10y^2, 15x^2 - 30xy + 15y^2, x^2 + 3xy + 2y^2$

EXAMPLE 3 Add Rational Expressions With Unlike Denominators

What is the sum of $\frac{x + 3}{x^2 - 1}$ and $\frac{2}{x^2 - 3x + 2}$?

Follow a similar procedure to the one you use to add numerical fractions with unlike denominators.

MAKE SENSE AND PERSEVERE
We must keep equivalent addends. So multiply each addend by a form of one. Choose the missing factors of the LCD.

$$\frac{x + 3}{x^2 - 1} + \frac{2}{x^2 - 3x + 2} = \frac{x + 3}{(x + 1)(x - 1)} + \frac{2}{(x - 1)(x - 2)}$$
Factor each denominator.
$$= \frac{(x + 3)(x - 2)}{(x + 1)(x - 1)(x - 2)} + \frac{2(x + 1)}{(x + 1)(x - 1)(x - 2)}$$
Use the LCM as the least common denominator (LCD).
$$= \frac{(x + 3)(x - 2) + 2(x + 1)}{(x + 1)(x - 1)(x - 2)}$$
$$= \frac{(x^2 + x - 6) + (2x + 2)}{(x + 1)(x - 1)(x - 2)}$$
In the numerator only, multiply factors, add like terms and factor again.
$$= \frac{x^2 + 3x - 4}{(x + 1)(x - 1)(x - 2)}$$
$$= \frac{(x + 4)(x - 1)}{(x + 1)(x - 1)(x - 2)}$$
$$= \frac{(x + 4)}{(x + 1)(x - 2)} \bullet \frac{(x - 1)}{(x - 1)}$$
Simplify and state the domain.
$$= \frac{x + 4}{(x + 1)(x - 2)} \text{ for } x \neq -1, 1, \text{ and } 2$$

The sum of $\frac{x + 3}{x^2 - 1}$ and $\frac{2}{x^2 - 3x + 2}$ is $\frac{x + 4}{(x + 1)(x - 2)}$ for $x \neq -1$, 1, and 2.

Try It! 3. Find the sum.
a. $\frac{x + 6}{x^2 - 4} + \frac{2}{x^2 - 5x + 6}$ b. $\frac{2x}{3x + 4} + \frac{4x^2 - 11x - 12}{6x^2 + 5x - 4}$

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RtI Support Struggling Students

USE WITH EXAMPLE 2 Students may have difficulty finding the LCM of polynomials.

Have students practice finding the LCM of integers.

1. Find the LCM of 24 and 42. [168]
2. Find the LCM of 11 and 17. [187]
3. Find the LCM of 15, 25, and 40. [600]

RtI Extend Student Thinking

USE WITH EXAMPLE 3 Challenge students by having them add three rational expressions with unlike denominators.

Find the sum.

1. $\dfrac{3}{x^2 - 4x + 3} + \dfrac{x + 4}{x^2 - 1} + \dfrac{x}{x^2 - 2x - 3}$ $\left[\dfrac{2x^2 + 3x - 9}{(x - 1)(x - 3)(x + 1)}\right]$

2. $\dfrac{x - 5}{x^2 + 6x + 8} + \dfrac{3 - x}{x^2 - 3x - 10} + \dfrac{4x + 1}{x^2 - x - 20}$ $\left[\dfrac{4x^2 - 2x + 39}{(x + 2)(x + 4)(x - 5)}\right]$

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Activity

EXAMPLE 4 Subtract Rational Expressions

Pose Purposeful Questions ETP

Q: How is subtracting rational expressions similar to adding rational expressions?

[When adding or subtracting, you need to find the least common denominator and rewrite the rational expressions before performing the operation.]

Q: How is the Distributive Property used when subtracting rational expressions?

[The Distributive Property is used when simplifying the numerator by subtracting polynomials. You multiply each term of the polynomial by -1.]

Try It! Answers

4. a. $\frac{(x-2)}{2x^2}$ for $x \neq 0$.

b. $\frac{1}{x-5}$ for $x \neq -5$ and 5.

Common Error

Try It! 4b When subtracting rational expressions, students may forget to distribute the negative sign. Have students relate subtracting rational expressions to subtracting polynomials, reminding them that they need to multiply each term by -1 before removing the parentheses and combining like terms in the numerator.

HABITS OF MIND

Use with **EXAMPLES 2, 3, & 4**

Communicate Precisely How does finding the LCM of two or more polynomials help you to add and subtract rational expressions?

[Once you know the LCM of polynomials in the denominators, you can multiply and divide each rational expression by the expression required to make each denominator the LCM. Then you can add or subtract the rational expressions.]

EXAMPLE 5 Use Rational Expressions to Solve Problems

Pose Purposeful Questions ETP

Q: How do you know that you are supposed to add the two rational expressions?

[The problem asks you to find an expression for the total time. Each expression represents the time for part of the trip. You need to add the times to find the total time.]

Q: How is the table a useful tool when solving this problem?

[The table organizes all of the information in the problem statement and the picture in one place. From the table, it is easy to see the expressions needed to calculate the total time.]

Try It! Answer

5. $\frac{40r + 100}{r(r+5)}$ or $\frac{20(2r+5)}{r(r+5)}$

Examples are available at SavvasRealize.com.

EXAMPLE 4 Subtract Rational Expressions

What is the difference between $\frac{x+1}{x^2-6x-16}$ and $\frac{x+1}{x^2+6x+8}$?

$$\frac{x+1}{x^2-6x-16} - \frac{x+1}{x^2+6x+8} = \frac{x+1}{(x-8)(x+2)} - \frac{x+1}{(x+2)(x+4)}$$

The LCD is $(x-8)(x+2)(x+4)$.

$$= \frac{(x+1)(x+4)}{(x-8)(x+2)(x+4)} - \frac{(x-8)(x+1)}{(x-8)(x+2)(x+4)}$$

$$= \frac{(x^2+5x+4)-(x^2-7x-8)}{(x-8)(x+2)(x+4)}$$

$$= \frac{x^2+5x+4-x^2+7x+8}{(x-8)(x+2)(x+4)}$$

Check for common factors in the numerator and denominator and simplify, if possible.

$$= \frac{12x+12}{(x-8)(x+2)(x+4)}$$

$$= \frac{12(x+1)}{(x-8)(x+2)(x+4)}$$

COMMON ERROR When subtracting polynomials, remember to distribute -1 when removing the parentheses.

The difference between $\frac{x+1}{x^2-6x-16}$ and $\frac{x+1}{x^2+6x+8}$ is $\frac{12(x+1)}{(x-8)(x+2)(x+4)}$ for $x \neq -4, -2$, and 8.

Try It! 4. Simplify.

a. $\frac{1}{3x} + \frac{1}{6x} - \frac{1}{x^2}$

b. $\frac{3x-5}{x^2-25} - \frac{2}{x+5}$

APPLICATION

EXAMPLE 5 Use Rational Expressions to Solve Problems

Jorge drives his car to the mechanic, then he takes the commuter rail train back to his neighborhood. The average speed for the 10-mile trip is 15 miles per hour faster on the train. Find an expression for Jorge's total travel time. If he drove 30 mph, how long did this take?

USE APPROPRIATE TOOLS Use a table to organize information and help create an accurate model of the situation.

	Distance	Rate	Time
Car	10	r	$\frac{10}{r}$
Commuter Rail	10	$r + 15$	$\frac{10}{r+15}$

Remember: distance = rate • time, so time = $\frac{\text{distance}}{\text{rate}}$.

Total time for the trip:

$$\frac{10}{r} + \frac{10}{r+15} = \frac{10(r+15)}{r(r+15)} + \frac{10r}{r(r+15)}$$

Add the times for each part of the trip.

$$= \frac{10r + 150 + 10r}{r(r+15)}$$

$$= \frac{20r + 150}{r(r+15)}$$

CONTINUED ON THE NEXT PAGE

HABITS OF MIND

Use with **EXAMPLE 5**

Construct Arguments Does Jorge spend more time traveling to his friend's house, or back home? Support your answer algebraically.

[traveling to his friend's house. You can express the time traveled to Jorge's friend's house by $\frac{20}{r}$ and the time home by $\frac{20}{r+5}$. Since $\frac{20}{r+5} < \frac{20}{r}$, Jorge spends less time traveling home.]

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Activity

EXAMPLE 6 Simplify a Compound Fraction

Examples are available at SavvasRealize.com.

Pose Purposeful Questions ETP

Q: What do you notice about a compound fraction that is different from other fractions?
[The numerator or denominator of a compound fraction contains a fraction.]

Q: Is there another method you could use to write a simpler form of the compound fraction?
[You can multiply the numerator by the reciprocal of the denominator, then write the rational expression as a single fraction.]

Try It! Answers

6. a. $\frac{3}{x^2 + 11}$ for $x \neq 1$. b. $\frac{2x - 1}{x^2 + 2}$ for $x \neq 0$.

Elicit and Use Evidence of Student Thinking ETP

Q: What first step would you take to rewrite the compound fraction?
[The first step is to multiply the numerator and denominator by the LCD of all rational expressions in the numerator and denominator. This eliminates the fractions from the numerator and denominator and makes the compound fraction easier to simplify.]

HABITS OF MIND

Use with EXAMPLE 6

Reason Edwin multiplied the top and bottom of the fraction in Try It 6a by $\frac{3}{x+1} + \frac{x-1}{4}$. Will this technique work to simplify the compound fraction? Explain.

[No; this will not simplify the denominator because the product of $\frac{x+1}{3} + \frac{4}{x-1}$ and $\frac{3}{x+1} + \frac{x-1}{4}$ gives 2 plus additional terms.]

EXAMPLE 5 CONTINUED

At a driving rate of 30 mph, you can find the total time.

$$\frac{20r + 150}{r(r + 15)} = \frac{20(30) + 150}{30(30 + 15)}$$

substitute 30 mph for the rate, and simplify.

$$= \frac{750}{1{,}350}$$

$$= \frac{5}{9}$$

The expression for Jorge's total travel time is $\frac{20r + 150}{r(r + 15)}$. The total time is $\frac{5}{9}$ h, or about 33 min.

Try It! 5. Jorge drives to visit his friends. His 20 mile trip to their house is congested with traffic, but it has cleared by the time he travels home. He can travel 5 miles per hour faster. Write an expression that represents Jorge's total travel time.

EXAMPLE 6 Simplify a Compound Fraction

VOCABULARY
A **compound fraction** contains a rational expression in its numerator, denominator, or both.

Write a simpler form of the compound fraction $\frac{\frac{1}{x} + \frac{2}{x+1}}{\frac{1}{y}}$.

Find the Least Common Multiple (LCM) of all the denominators.

$$\frac{\frac{1}{x} + \frac{2}{x+1}}{\frac{1}{y}} = \frac{\left[\frac{1}{x} + \frac{2}{x+1}\right] \cdot \frac{[x(x+1)y]}{1}}{\frac{1}{y} \cdot \frac{[x(x+1)y]}{1}}$$

Multiply the fractions in the numerator and the denominator by the LCM over 1.

$$= \frac{\frac{1}{\cancel{x}} \cdot \frac{\cancel{x}(x+1)y}{1} + \frac{2}{\cancel{x+1}} \cdot \frac{x\cancel{(x+1)}y}{1}}{\frac{1}{\cancel{y}} \cdot \frac{x(x+1)\cancel{y}}{1}}$$

$$= \frac{\frac{(x+1)y}{1} + \frac{2xy}{1}}{\frac{x(x+1)}{1}}$$

The individual fractions now have denominators equal to 1.

$$= \frac{(x+1)y + 2xy}{x(x+1)}$$

A fraction with 1 in the denominator may be written without the 1.

$$= \frac{xy + y + 2xy}{x(x+1)}$$

$$= \frac{(3x+1)y}{x(x+1)}$$

So, $\frac{\frac{1}{x} + \frac{2}{x+1}}{\frac{1}{y}}$ is equivalent to $\frac{(3x+1)y}{x(x+1)}$ when $x \neq -1$, 0 and $y \neq 0$.

Try It! 6. Simplify each compound fraction.

a. $\frac{\frac{1}{x-1}}{\frac{x+1}{3} + \frac{4}{x-1}}$ b. $\frac{2 - \frac{1}{x}}{x + \frac{2}{x}}$

208 TOPIC 4 Rational Functions

ELL English Language Learners (*Use with* EXAMPLE 6)

READING BEGINNING Help students understand the steps in the method by previewing the math vocabulary before reading the example. Show the students the following terms and their definitions: *LCD*, *numerator*, *denominator*, and *reciprocal*. Give students a copy of the math steps in the example. Have them read the example and label the terms as they are reading.

Q: How is the LCD different from the denominator in the compound fraction?
The LCD is the product of all of the denominators that appear in the compound fraction x, y, and $x + 1$. The denominator of the compound fraction is $\frac{1}{y}$.

SPEAKING DEVELOPING As you review the steps in simplifying a compound fraction, discuss the terms *eliminate* and *reciprocal*. Have students take turns explaining the *process* in each method.

Q: When multiplying the numerator and denominator of the compound fraction by the LCD, what is being eliminated?
[Multiplying the numerator and denominator by the LCD "eliminates" fractions. The compound fraction is rewritten with a numerator and denominator with no fractions.]

Q: What property is helpful in eliminating the fractions?
[The Distributive Property is helpful for multiplying each addend in the brackets by the LCD.]

LISTENING EXPANDING Have students work in groups. Give each group cards labeled with numbers, operations, and fractions. Read aloud the problem statement from the example about a *compound fraction*. As they listen to the description, have students arrange the cards to create a compound fraction.

Q: Does a compound fraction have to include more than one fraction?
[No; the description says *one or more fractions*.]

Q: Does a compound fraction have to have fractions in both the numerator and denominator?
[No; the definition says *and/or*, which means a compound fraction may have a fraction in either the numerator or the denominator, or it may have fractions in both.]

CONCEPT SUMMARY Find the Sums and Differences of Rational Expressions

Q: Are the steps for subtracting two rational expressions the same as the steps for adding two rational expressions? [When subtracting two rational expressions, you must remember to distribute the negative sign to each term in the numerator of the expression being subtracted.]

Do You UNDERSTAND? | Do You KNOW HOW?

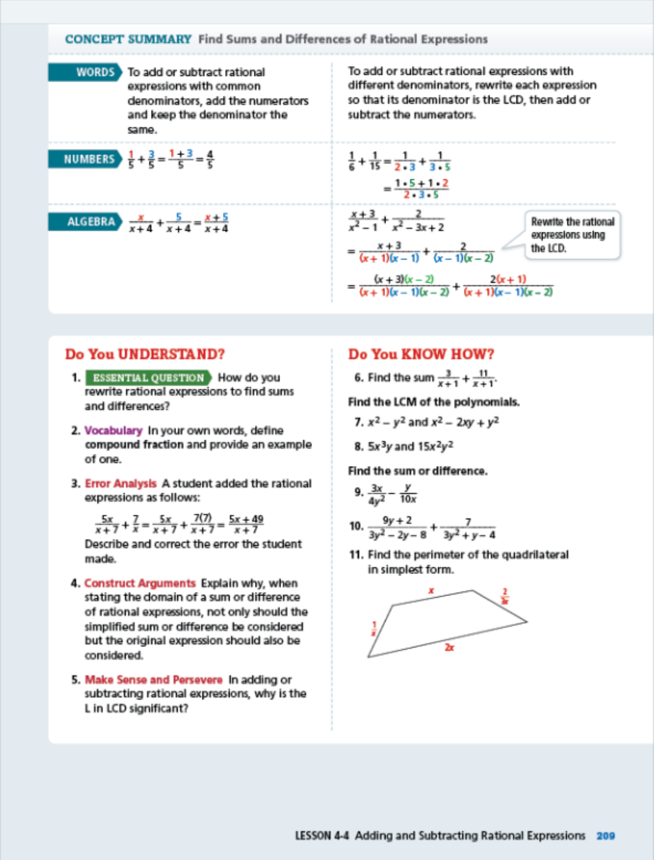
Common Error

Exercise 7 Students may incorrectly find the LCM by multiplying the two polynomials. Have students factor each polynomial and identify any repeated factors before they multiply to find the LCM.

Answers

1. Find a common denominator, add or subtract the numerators, and simplify, eliminating common factor(s) shared by both the numerator and denominator of the sum or difference.
2. A compound fraction is a fraction that has one or more fractions in the numerator and/or denominator.
$$\frac{\frac{1}{2}}{\frac{2x}{x+y}}$$
3. The student incorrectly used the polynomial expression $x + 7$ as the common denominator, when $x(x + 7)$ should have been used. The correct method is $\frac{5x}{x+7} + \frac{7}{x} = \frac{5x(x)}{x(x+7)} + \frac{7(x+7)}{x(x+7)} = \frac{5x^2+7x+49}{x(x+7)}$.
4. The simplified sum or difference may not include some values of the domain that were eliminated during the simplification process.
5. LCD stands for *least common denominator.* The sum or difference will not be in simplest form if you multiply the numerator and denominator of each expression by more factors than necessary.
6. $\frac{14}{x+1}$
7. $(x + y)(x - y)^2$
8. $15x^3y^2$
9. $\frac{15x^2 - 2y^3}{20xy^2}$
10. $\frac{3y - 4}{(y - 2)(y - 1)}$
11. $\frac{9x^2 + 5}{3x}$ units

Concept Summary, Do You Understand, and **Do You Know How** are available at SavvasRealize.com.



CONCEPT SUMMARY Find Sums and Differences of Rational Expressions

WORDS	To add or subtract rational expressions with common denominators, add the numerators and keep the denominator the same.	To add or subtract rational expressions with different denominators, rewrite each expression so that its denominator is the LCD, then add or subtract the numerators.
NUMBERS	$\frac{1}{5}+\frac{3}{5}=\frac{1+3}{5}=\frac{4}{5}$	$\frac{1}{6}+\frac{1}{15}=\frac{1}{2\cdot3}+\frac{1}{3\cdot5}$ $=\frac{1\cdot5+1\cdot2}{2\cdot3\cdot5}$
ALGEBRA	$\frac{x}{x+4}+\frac{5}{x+4}=\frac{x+5}{x+4}$	$\frac{x+3}{x^2-1}+\frac{2}{x^2-3x+2}$ $=\frac{x+3}{(x+1)(x-1)}+\frac{2}{(x-1)(x-2)}$ $=\frac{(x+3)(x-2)}{(x+1)(x-1)(x-2)}+\frac{2(x+1)}{(x+1)(x-1)(x-2)}$ Rewrite the rational expressions using the LCD.

Do You UNDERSTAND?

1. ESSENTIAL QUESTION How do you rewrite rational expressions to find sums and differences?
2. Vocabulary In your own words, define compound fraction and provide an example of one.
3. Error Analysis A student added the rational expressions as follows:
$$\frac{5x}{x+7}+\frac{7}{x}=\frac{5x}{x+7}+\frac{7(7)}{x+7}=\frac{5x+49}{x+7}$$
Describe and correct the error the student made.
4. Construct Arguments Explain why, when stating the domain of a sum or difference of rational expressions, not only should the simplified sum or difference be considered but the original expression should also be considered.
5. Make Sense and Persevere In adding or subtracting rational expressions, why is the L in LCD significant?

Do You KNOW HOW?

6. Find the sum $\frac{3}{x+1}+\frac{11}{x+1}$.

Find the LCM of the polynomials.

7. $x^2 - y^2$ and $x^2 - 2xy + y^2$
8. $5x^3y$ and $15x^2y^2$

Find the sum or difference.

9. $\frac{3x}{4y^2}-\frac{y}{10x}$
10. $\frac{9y+2}{3y^2-2y-8}+\frac{7}{3y^2+y-4}$
11. Find the perimeter of the quadrilateral in simplest form.

x $\frac{2}{3x}$ $\frac{1}{x}$ $2x$

LESSON 4-4 Adding and Subtracting Rational Expressions 209

PRACTICE & PROBLEM SOLVING

Practice & Problem Solving and **video tutorials** are available at SavvasRealize.com.

Lesson Practice You may opt to have students complete the automatically scored Practice and Problem Solving items online powered by MathXL for School.

Choose from: **Lesson Practice**

Adaptive Practice

You may also take advantage of the bank of exercises for assigning additional practice.

Assignment Guide

On-Level	Advanced
15, 16, 18–31, 34–36	12–17, 22–36

Engage Through Student Choice

Promote student agency by allowing students to choose practice items. You may structure this choice in many ways.

For example:

Assign each section a point value. Students choose at least one item from each section and items chosen should have a minimum of 20 total points.

Understand, Apply....................2 points each

Practice....................................1 point each

Assessment Practice.................1 point each

Performance Task ....3 points each

Example	Items	DOK
1	18-19	1
	15	2
2	20-21	1
	16	2
3	22–23	1
	13, 17	2
4	24, 25, 35	1
	12	2
5	26	2
	31, 33	3
6	27–30	2
	14, 32, 34	3

Scan the QR code to access a video tutorial.

PRACTICE & PROBLEM SOLVING

SavvasRealize.com

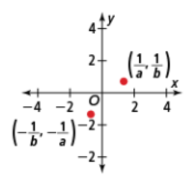
UNDERSTAND

12. **Generalize** Explain how addition and subtraction of rational expressions is similar to and different from addition and subtraction of rational numbers.

13. **Error Analysis** Describe and correct the error a student made in adding the rational expressions.

$$\begin{aligned}\frac{1}{x^2+3x+2}+\frac{x^2+4x}{4x+8} &= \frac{1}{(x+1)(x+2)}+\frac{x(x+4)}{4(x+2)}\\ &= \frac{4}{4(x+1)(x+2)}+\frac{x(x+4)}{4(x+1)(x+2)}\\ &= \frac{4+x^2+4x}{4(x+1)(x+2)}\\ &= \frac{x^2+4x+4}{4(x+1)(x+2)}\\ &= \frac{(x+2)(x+2)}{4(x+1)(x+2)}\\ &= \frac{x+2}{4(x+1)}\cdot\frac{(x+2)}{(x+2)}\\ &= \frac{x+2}{4(x+1)} \quad ✗\end{aligned}$$

14. **Use Structure** Find the slope of the line that passes through the points shown. Express in simplest form. 1



15. **Reason** For what values of x is the sum of $\frac{x-5y}{x+y}$ and $\frac{x+7y}{x+y}$ undefined? Explain.

16. **Look for Relationships** Find the LCM of $3x^2+7x+2$ and $9x+3$.

17. **Higher Order Thinking** Why are rational expressions closed under addition?

PRACTICE

Find the sum. SEE EXAMPLE 1

18. $\frac{4x}{x+7}+\frac{9}{x+7}$

19. $\frac{3y-1}{y^2+4y}+\frac{9y+6}{y(y+4)}$

Find the LCM for each group of expressions. SEE EXAMPLE 2

20. $x^2-7x+6,\ x^2-5x-6$ $(x-6)(x+1)(x-1)$

21. $y^2+2y-24,\ y^2-16,\ 2y$ $2y(y+6)(y+4)(y-4)$

Find the sum. SEE EXAMPLE 3

22. $\frac{6x}{x^2-8x}+\frac{4}{2x-16}$

23. $\frac{3y}{2y^2-y}+\frac{2}{2y}$

Find the difference. SEE EXAMPLE 4

24. $\frac{4x}{x^2-1}-\frac{4}{x-1}$

25. $\frac{y-1}{3y+15}-\frac{y+3}{5y+25}$

26. On Saturday morning, Ahmed decided to take a bike ride from one end of the 15-mile bike trail to the other end of the bike trail and back. His average speed the first half of the ride was 2 mph faster than his speed on the second half. Find an expression for Ahmed's total travel time. If his average speed for the first half of the ride was 12 mph, how long was Ahmed's bike ride? SEE EXAMPLE 5

Write each expression in its simplest form. SEE EXAMPLE 6

27. $\dfrac{1+\frac{1}{x}}{x-\frac{1}{x}}$

28. $\dfrac{\frac{3}{y}+\frac{7}{x}}{\frac{1}{y}-\frac{2}{x}}$

29. $\dfrac{\frac{1}{a}+\frac{1}{b}}{\frac{a^2-b^2}{ab}}$

30. $\dfrac{\frac{z^2-z-12}{z^2-2z-15}}{\frac{z^2+8z+12}{z^2-5z-14}}$

210 TOPIC 4 Rational Functions

Answers

12. The same basic procedures are followed for both. First, find a common denominator, then rewrite the expressions using the common denominator, perform the addition and/or subtraction, and then simplify.

13. When rewriting the second expression with a common denominator, the student should have had $x(x+1)(x+4)$ in the numerator rather than $x(x+4)$.

15. When $x=-y$, the denominator is zero, which is undefined.

16. $3(3x+1)(x+2)$

See answers for Exercises 17–19, 22–30 on next page.

Answers

17. You can find the sum of two rational expressions $\frac{p(x)}{q(x)} + \frac{r(x)}{t(x)}$ by finding the sum of equivalent expressions,

$$\frac{p(x)\cdot t(x)}{(q(x)\cdot t(x)} + \frac{r(x)\cdot q(x)}{t(x)\cdot q(x)} = \frac{(p(x)\cdot t(x) + r(x)\cdot q(x)}{q(x)\cdot t(x)}.$$

Since the numerator and denominator are still polynomials, the sum is a rational expression. So rational expressions are closed under addition.

18. $\frac{4x+9}{x+7}$

19. $\frac{12y+5}{y^2+4y}$

22. $\frac{8}{x-8}$

23. $\frac{5y-1}{y(2y-1)}$

24. $\frac{-4}{(x+1)(x-1)}$

25. $\frac{2y-14}{15(y+5)}$ or $\frac{2(y-7)}{15(y+5)}$

26. $\frac{30(x+1)}{x(x+2)}$; 2.75 hours

27. $\frac{1}{x-1}$

28. $\frac{3x+7y}{x-2y}$

29. $\frac{1}{a-b}$

30. $\frac{(z-4)(z-7)}{(z-5)(z+6)}$

31. a. $\frac{9x+7}{(x+3)(x-1)}$

 b. 5 h

32. $(x-5)(x+4) : 1$ or $x^2 - x - 20 : 1$

33. a. $\frac{180x+720}{x(x+8)}$ or $\frac{180(x+4)}{x(x+8)}$

 b. ≈ 3.1 h

36. PART A $f = 4$ cm

 PART B $x \approx 3.2$ inches

Practice & Problem Solving is available at SavvasRealize.com.

PRACTICE & PROBLEM SOLVING

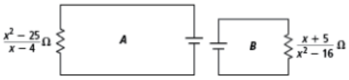
APPLY

31. **Use Structure** Aisha paddles a kayak 5 miles downstream at a rate 3 mph faster than the river's current. She then travels 4 miles back upstream at a rate 1 mph slower than the river's current. Hint: Let x represent the rate of the river current.

 a. Write and simplify an expression to represent the total time it takes Aisha to paddle the kayak 5 miles downstream and 4 miles upstream.

 b. If the rate of the river current, x, is 2 mph, how long was Aisha's entire kayak trip?

32. **Model With Mathematics** The resistance of electrical circuits A and B are shown. Find the ratio of the resistance of Circuit A to Circuit B.



33. **Reason** The Bello family drives 180 miles (round trip) to a professional basketball game. On the way to the game, their average speed is approximately 8 mph faster than their speed on the return trip home.

 a. Let x represent their average speed on the way home. Write and simplify an expression to represent the total time it took them to drive to and from the game.

 b. If their average speed going to the game was 62 mph, how long did it take them to drive to the game and back?

ASSESSMENT PRACTICE

34. Which of the following compound fractions simplifies to $\frac{x+1}{x-3}$? Select all that apply.

- ■ $\dfrac{\frac{x^2+5x+4}{x^2+2x-8}}{\frac{x^2-4x+3}{x^2-3x+2}}$
- □ $\dfrac{\frac{x^2-1}{x^2-4}}{\frac{x^2+x-7}{x^2+5x+6}}$
- □ $\dfrac{\frac{x^2+3x-10}{x^2-16}}{\frac{x^2-4x-5}{x^2-1}}$
- ■ $\dfrac{\frac{x^2+3x-10}{x^2-5x+6}}{\frac{x^2-25}{x^2-4x-5}}$

35. **SAT/ACT** What is the difference between $\frac{x}{9}$ and $\frac{x-y}{6}$?

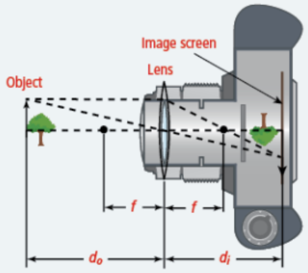
Ⓐ $\frac{5x-y}{18}$

Ⓑ $\frac{5x+y}{18}$

● $\frac{-x+3y}{18}$

Ⓓ $\frac{-x-3y}{18}$

36. **Performance Task** The lens equation $\frac{1}{f} = \frac{1}{d_i} + \frac{1}{d_o}$ represents the relationship between f, the focal length of a camera lens, d_i, the distance from the lens to the film, and d_o, the distance from the lens to the object.



Part A Find the focal length of a camera lens if an object that is 12 cm from a camera lens is in focus on the film when the lens is 6 cm from the film.

Part B Suppose the focal length of another camera lens is 3 inches, and the object to be photographed is 5 feet away. What distance (to the nearest tenth inch) should the lens be from the image screen?

LESSON QUIZ

Use the Lesson Quiz to assess students' understanding of the mathematics in the lesson.

Students can take the Lesson Quiz online or you can download a printable copy from **SavvasRealize.com.** The Lesson Quiz is also available in the *Assessment Sourcebook*.

Item Analysis

Item	DOK	Skills Review and Practice
1	1	A86
2	2	A86
3	1	A86
4	2	A86
5	2	A86

RtI Use the student scores on the Lesson Quiz to prescribe differentiated assignments.

If students take the Lesson Quiz online, it will be automatically scored and appropriate differentiated practice will be assigned based on student performance.

I Intervention	0–3 points	• Reteach to Build Understanding • Mathematical Literacy and Vocabulary • Lesson Virtual Nerd videos
O On-Level	4 points	• Enrichment
A Advanced	5 points	• Enrichment

Lesson Quiz is available at SavvasRealize.com.

ASSESSMENT RESOURCES

Name ______________________ enVision Algebra2 SavvasRealize.com

4-4 Lesson Quiz

Adding and Subtracting Rational Expressions

1. Add $\frac{5}{2x+1} + \frac{8}{4x+2}$, with $x \neq -\frac{1}{2}$.
   - Ⓐ $\frac{9}{2x+1}$
   - Ⓑ $\frac{13}{4x+2}$
   - Ⓒ $\frac{5x+8}{4x+2}$
   - Ⓓ $\frac{18}{2x+1}$
2. What is the LCM for $(x+3)^2(x-2)$ and $(x+3)(x^2-16)$?
   - Ⓐ $x+3$
   - Ⓑ $(x+3)(x-2)(x-16)$
   - Ⓒ $(x+3)^3(x-2)(x^2+16)$
   - Ⓓ $(x+3)^2(x-2)(x^2-16)$
3. Subtract $\frac{3}{x-1} - \frac{4}{x+3}$.
   - Ⓐ $\frac{-x+13}{(x-1)(x+3)}$; $x \neq -3$ or 1
   - Ⓑ $\frac{-x+5}{(x-1)(x+3)}$; $x \neq -3$ or 1
   - Ⓒ $\frac{7x+5}{(x-1)(x+3)}$; $x \neq -3$ or 1
   - Ⓓ $\frac{7x+13}{(x-1)(x+3)}$; $x \neq -3$ or 1
4. Simplify $\frac{\frac{1}{x} - \frac{1}{x+2}}{2}$.
   - Ⓐ $\frac{1}{x+2}$; $x \neq -2$
   - Ⓑ $\frac{1}{x}$; $x \neq 0$
   - Ⓒ $\frac{1}{x(x+2)}$; $x \neq -2$ or 0
   - Ⓓ $\frac{1}{x(x-2)}$; $x \neq 2$ or 0
5. The equation $r = \frac{1}{\frac{1}{r_1} + \frac{1}{r_2}}$ represents the total resistance, r, when two resistors whose resistances are r_1 and r_2 are connected in parallel. Find the total resistance when r_1 is x and r_2 is $x+1$.
   - Ⓐ $\frac{1}{2x+1}$; $x \neq -1, -\frac{1}{2}$, or 0
   - Ⓑ $2x+1$; $x \neq -1$ or 0
   - Ⓒ $\frac{x(x+1)}{2x+1}$; $x \neq -1, -\frac{1}{2}$, or 0
   - Ⓓ $\frac{2x+1}{x(x+1)}$; $x \neq -1$ or 0

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Practice

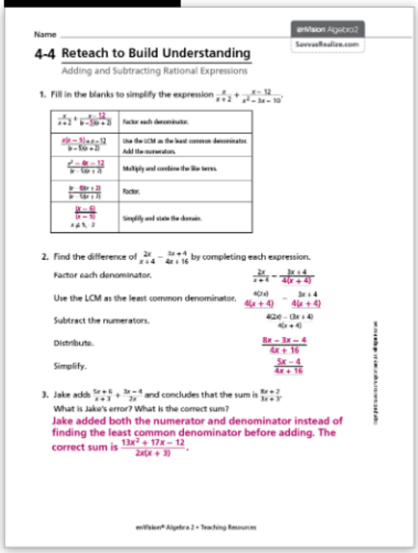
DIFFERENTIATED RESOURCES

I = Intervention O = On-Level A = Advanced

Reteach to Build Understanding I

Provides scaffolded reteaching for the key lesson concepts.

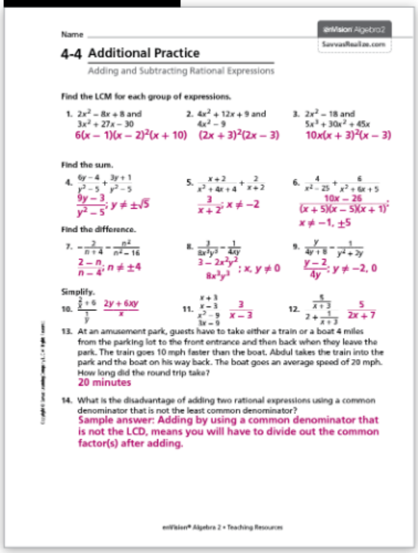
MathXL® for School



Additional Practice I O

Provides extra practice for each lesson.

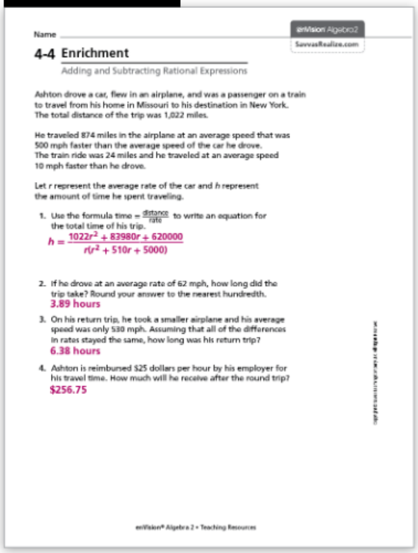
MathXL® for School



Enrichment O A

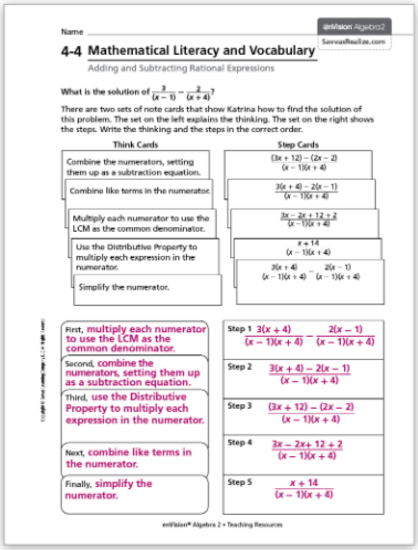
Presents engaging problems and activities that extend the lesson concepts.

MathXL® for School



Mathematical Literacy and Vocabulary I O

Helps students develop and reinforce understanding of key terms and concepts.



Digital Differentiated Resources

The **Reteach to Build Understanding**, **Additional Practice**, and **Enrichment** activities are available as digital assignments. These activities are automatically assigned when students complete the lesson quiz online and are automatically scored.